YU WANG

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EDUCATION

University of Glasgow *Glasgow, United Kingdom*
PhD in Geospatial Data Science *2022 – 2025*

ESIEE-Paris (École supérieure d'ingénieurs en électrotechnique et électronique) *Paris, France*
Diplôme d'Ingénieur eq. M.Sc. of Telecommunication, Score: 14.9/20 *2012 – 2014*

Harbin Institute of Technology *Harbin, China*
BEng Telecommunication Engineering *2007 – 2011*

RESEARCH EXPERIENCE

University of Glasgow Feb. 2025 – Present
Geospatial Artificial Intelligence Research Scientist *United Kingdom*

- Analysed spatial and temporal bias in large-scale street-view imagery by analysing coverage and update patterns, revealing systematic disparities affecting fairness and representation in GeoAI datasets.
- Contributed to the Consultancy AI project developing an AI-powered platform for automated legal document review, supporting the digitisation and efficiency of university consultancy contract workflows.
- Developed LLM- and RAG-based tools for automated extraction, review, and revision of legal clauses, improving the efficiency and consistency of contract analysis within the consultancy service.

DSO–Alan Turing Institute Apr. 2024 – Oct. 2024
Research Assistant *United Kingdom*

- Developed methods for extracting geographic location information from visual data to support geospatial AI and visual geolocation applications.
- Curated a large-scale global street-view imagery dataset to support geospatial AI research, ensuring geographic diversity and consistent data quality.

University of Glasgow Jan. 2022 – Dec. 2025
PhD in Geospatial Data Science *United Kingdom*

- Developed a novel framework for 3D urban modelling using cellular mobile signals, demonstrating the feasibility of opportunistic wireless sensing as a scalable alternative to traditional LiDAR-based mapping.
- Designed and implemented a prototype mobile application for large-scale real-world data collection, enabling opportunistic sensing for 3D city modelling and bridging geospatial analytics with mobile computing.
- Identified spatial and temporal bias in large-scale street-view imagery and quantified its impact on cross-view geolocalisation models, highlighting how uneven data coverage affects model robustness and the representation of urban environments in GeoAI systems.
- Advanced human activity recognition using ultra-wideband (UWB) channel impulse response data by benchmarking eleven classical and deep learning models, demonstrating the effectiveness of UWB sensing for fine-grained human behaviour recognition.
- Investigated key personal, situational, and environmental factors contributing to pedestrian disorientation in Greater London through a mixed-methods study, providing insights for human-centred urban design and navigation systems.

WORK EXPERIENCE

Bouygues Telecom Aug. 2017 – Jan. 2021
IP Network Performance Engineer *France*

- Led network capacity planning by developing network load growth forecasting models to project 3–5-year demand trends and optimise future network expansion strategies.
- Directed network performance analysis within the "Network Redesigning" project (since 2019), providing technical assessments and optimisation recommendations.
- Oversaw the departmental initiative "Automation of Weekly/Monthly KPI Reports", coordinating with network architecture and IT teams to ensure alignment with project objectives.
- Delivered monthly QoE/QoS performance reports for mobile, FTTH, and ADSL services, including data visualisation, analytical insights, and technical findings for departmental presentations.
- Managed infrastructure material resources by monitoring utilisation and prioritising allocation to meet operational needs.

Davidson Consulting Nov. 2014 – Aug. 2017
Research Engineer *France*

- Developed a network performance monitoring and usage-tracking framework to optimise decision-making and support continuous QoS improvement across long-term infrastructure projects.
- Collaborated with network deployment and implementation teams to manage upgrade lifecycles, from initial demand assessment to final rollout and performance validation.

Bouygues Telecom, Internship Apr. 2014 – Oct. 2014
Optical & Microwave Engineer *France*

- Conducted analysis and optimisation of capacity resources for microwave transmission links as part of a long-haul network performance project.
- Designed and implemented a simulation tool to support decision-making in replacing STM-1 channels with Ethernet links in the Fujitsu FRX-3 long-haul microwave radio system.

JOURNAL AND CONFERENCE PUBLICATIONS

- Wang, Y., & Basiri, A. (2025). Bit to brick: from cellular mobile signals to 3D city map creation. ***Big Earth Data***, 1–25.
- Wang, Y., & Basiri, A. (2024). Advancing human activity recognition using ultra-wideband channel impulse response snapshots. In *Proceedings of the 2024 International Conference on Activity and Behavior Computing (**ABC**)* (pp. 1–10). IEEE.
- Wang, Y., Basiri, A., Gerrits, P., Solomon, G., Woelk, S., & Fidel Pereira, M. (2026). Why do pedestrians get lost? A case study of personal, situational, and environmental factors in Greater London. ***Journal of Location Based Services***, 1–26.
- Wang, Y., Shingleton, J., Bıcakçı, Y. S., Basiri, A., & Ackland, J. (2026). The Invisible City: Not All Streets Are Seen Equally. In *Proceedings of the 1st International Conference on Geospatial Artificial Intelligence (**GeoAI 2026**)*, Ghent, Belgium. (Oral Presentation)
- Shingleton, J., Bıcakçı, Y. S., Wang, Y., & Basiri, A. (2026). Emergent spatio-semantic structure in large language model embedding spaces. In *Proceedings of the 1st International Conference on Geospatial Artificial Intelligence (**GeoAI 2026**)*, Ghent, Belgium. (Oral Presentation)

- Bıcakçı, Y. S., Shingleton, J., Wang, Y., & Basiri, A. (2026). Mapping the semantics of the street: A VLM-driven geospatial analysis in Fatih, Istanbul. In *Proceedings of the 1st International Conference on Geospatial Artificial Intelligence (**GeoAI 2026**)*, Ghent, Belgium. (Oral Presentation)
- Wang, Y., & Basiri, A. (under review). From static to dynamic: evaluating the impact of temporal bias in historical street view images for cross-view geolocalisation. ***Transactions in GIS***.
- Shingleton, J., Bıcakçı, Y. S., Wu, M., Wang, Y., & Basiri, A. (under review). Evaluating explanation integrity in a vision language model for urban perception. ***Transactions in GIS***.

AWARDS

- Excellent paper award (Top 2) - International Conference on Activity and Behavior Computing — Japan, 2024
- Conference Support Funding (£500) — UK, 2024
- Winner of the outstanding international graduates of ESIEE-Paris (sole recipient) — France, 2014

FUNDED RESEARCH PROJECTS

- **Consultancy AI Project**, University of Glasgow Consultancy Service. Research Assistant. Development of AI-assisted tools for automated legal contract analysis. 2025–Present
- **Geospatial AI Research Project**, DSO National Laboratories (Singapore). Research Assistant. Visual geolocation and geospatial data analysis. 2024

TECHNICAL SKILLS

- **Languages & Frameworks**: Python, MATLAB, Kotlin, computer vision, TensorFlow, PyTorch
- **Data analysis & visualisation**: Tableau, SAP Business Intelligence
- **Telecommunications**: Transmission networks, QoS, network performance, capacity planning

LICENSES & CERTIFICATIONS

- **Coursera Specialisations**: Machine Learning (DeepLearning.AI), Deep Learning (DeepLearning.AI & Stanford Online), Python (University of Michigan).
- **Coursera Short Courses**: Data Science Math Skills (Duke University), Matrix Methods (University of Minnesota).

LANGUAGES

- English (fluent)
- French (fluent)
- Mandarin (native)